

Project Initialization and Planning Phase

Date	28 JUNE 2027
Team ID	xxxxxxx
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to improve the credit card approval process using machine learning, enabling faster, more accurate, and consistent approval decisions. It addresses the limitations of manual evaluation by reducing processing time, minimizing errors, and supporting better decision-making. The system predicts whether a credit card application should be approved based on applicant information and credit history.

Project Overview	
Objective	The primary objective is to develop a machine learning-based credit card approval prediction system that accurately predicts whether a customer's credit card application should be approved or rejected based on applicant and credit history data.
Scope	The project focuses on collecting, preprocessing, and analyzing applicant and credit history data, training multiple machine learning models, selecting the best-performing model, and deploying it through a Flask web application for real-time prediction.
Problem Statement	
Description	Traditional credit card approval processes are often time-consuming and may involve manual verification, leading to delays and inconsistent decision-making. Banks require an efficient system that can quickly assess an applicant's eligibility based on multiple financial and personal factors.

Impact	Implementing a machine learning-based prediction system improves decision accuracy, reduces processing time, minimizes human errors, supports faster approval decisions, and enhances customer satisfaction.
Proposed Solution	
Approach	The proposed solution uses machine learning algorithms to analyze applicant information and credit history. After preprocessing the data, multiple classification models are trained and evaluated. The best-performing model (XGBoost) is selected and deployed using Flask to provide instant credit card approval predictions.
Key Features	- Machine learning-based credit card approval prediction. - Comparison of multiple classification algorithms.

	- Fast and accurate prediction using the XGBoost model. - User-friendly Flask web application for real-time prediction. - Efficient data preprocessing and feature engineering.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5, optional GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	10 GB or more

Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn ,xgboost
Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data		
Dataset	Source, size, format	Kaggle Credit Card Approval Dataset (Application Record.csv & Credit Record.csv), CSV format